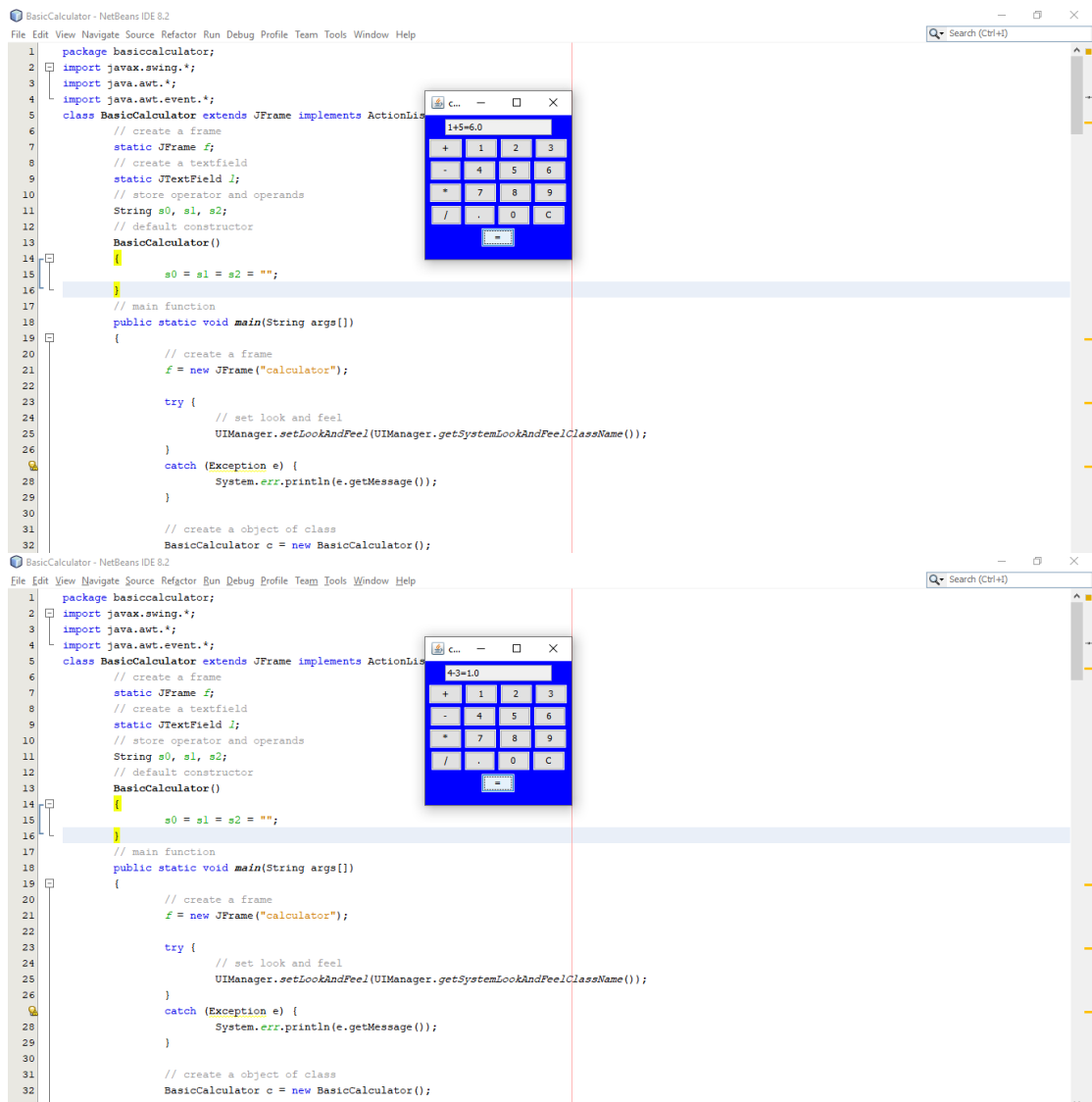
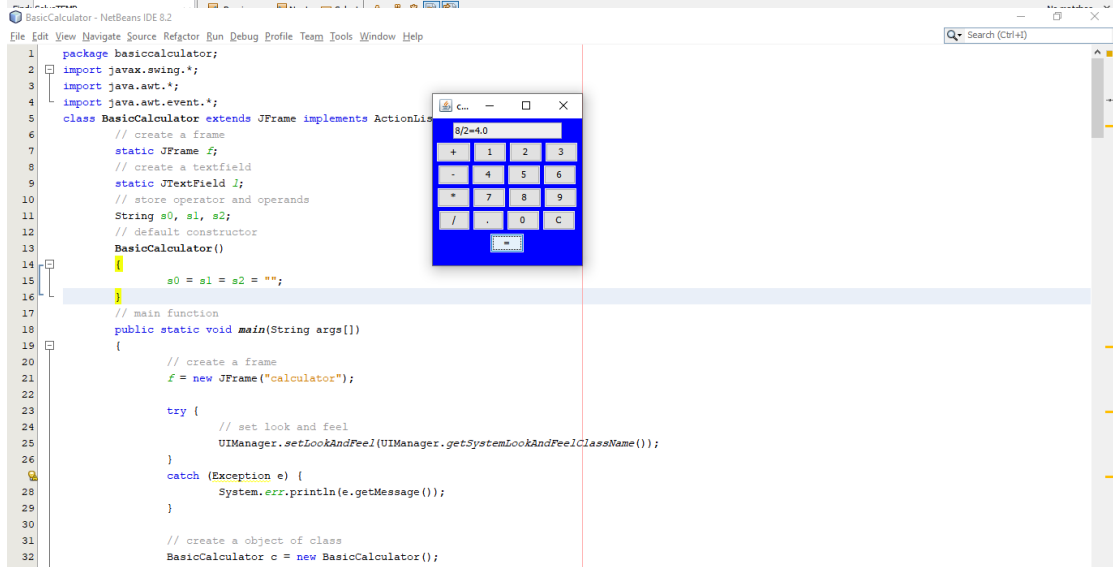
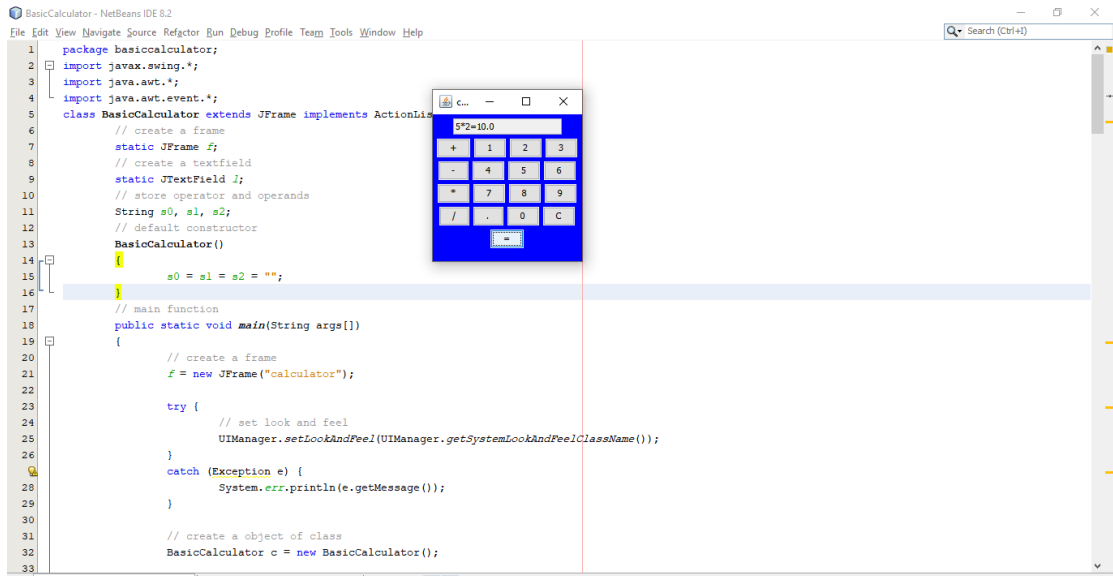


Create a Basic Calculator:

- Design a GUI frame with a text field for input, buttons for numbers and operations, and a display area for results.
- Implement event handling for button clicks to perform calculations and update the display.
- Explore different layout managers to organize the components effectively.





```

package basiccalculator;
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
class BasicCalculator extends JFrame implements
ActionListener {
    // create a frame
    static JFrame f;
    // create a textfield
    static JTextField l;
    // store operator and operands
    String s0, s1, s2;
    // default constructor
    BasicCalculator()
    {
        s0 = s1 = s2 = "";
    }
    // main function
    public static void main(String args[])
    {
        // create a frame
        f = new JFrame("calculator");

        try {
            // set look and feel

            UIManager.setLookAndFeel(UIManager.getSystemLook
AndFeelClassName());
        }
        catch (Exception e) {
            System.err.println(e.getMessage());
        }

        // create a object of class

```

```
BasicCalculator c = new BasicCalculator();

// create a textfield
l = new JTextField(16);

// set the textfield to non editable
l.setEditable(false);

// create number buttons and some operators
JButton b0, b1, b2, b3, b4, b5, b6, b7, b8, b9, ba, bs,
bd, bm, be, beq, beq1;

// create number buttons
b0 = new JButton("0");
b1 = new JButton("1");
b2 = new JButton("2");
b3 = new JButton("3");
b4 = new JButton("4");
b5 = new JButton("5");
b6 = new JButton("6");
b7 = new JButton("7");
b8 = new JButton("8");
b9 = new JButton("9");

// equals button
beq1 = new JButton("=");

// create operator buttons
ba = new JButton("+");
bs = new JButton("-");
bd = new JButton("/");
bm = new JButton("*");
beq = new JButton("C");

// create . button
be = new JButton(".");
```

```
// create a panel
JPanel p = new JPanel();

// add action listeners
bm.addActionListener(c);
bd.addActionListener(c);
bs.addActionListener(c);
ba.addActionListener(c);
b9.addActionListener(c);
b8.addActionListener(c);
b7.addActionListener(c);
b6.addActionListener(c);
b5.addActionListener(c);
b4.addActionListener(c);
b3.addActionListener(c);
b2.addActionListener(c);
b1.addActionListener(c);
b0.addActionListener(c);
be.addActionListener(c);
beq.addActionListener(c);
beq1.addActionListener(c);

// add elements to panel
p.add(l);
p.add(ba);
p.add(b1);
p.add(b2);
p.add(b3);
p.add(bs);
p.add(b4);
p.add(b5);
p.add(b6);
p.add(bm);
p.add(b7);
p.add(b8);
p.add(b9);
p.add(bd);
```

```

        p.add(be);
        p.add(b0);
        p.add(beq);
        p.add(beq1);

        // set Background of panel
        p.setBackground(Color.blue);

        // add panel to frame
        f.add(p);

        f.setSize(200, 220);
        f.show();
    }
    public void actionPerformed(ActionEvent e)
    {
        String s = e.getActionCommand();

        // if the value is a number
        if ((s.charAt(0) >= '0' && s.charAt(0) <= '9') ||
s.charAt(0) == '.') {
            // if operand is present then add to second no
            if (!s1.equals(""))
                s2 = s2 + s;
            else
                s0 = s0 + s;

            // set the value of text
            l.setText(s0 + s1 + s2);
        }
        else if (s.charAt(0) == 'C') {
            // clear the one letter
            s0 = s1 = s2 = "";

            // set the value of text
            l.setText(s0 + s1 + s2);
        }
    }

```

```

else if (s.charAt(0) == '=') {

    double te;

    // store the value in 1st
    if (s1.equals("+"))
        te = (Double.parseDouble(s0) +
Double.parseDouble(s2));
    else if (s1.equals("-"))
        te = (Double.parseDouble(s0) -
Double.parseDouble(s2));
    else if (s1.equals("/"))
        te = (Double.parseDouble(s0) /
Double.parseDouble(s2));
    else
        te = (Double.parseDouble(s0) *
Double.parseDouble(s2));

    // set the value of text
    l.setText(s0 + s1 + s2 + "=" + te);

    // convert it to string
    s0 = Double.toString(te);

    s1 = s2 = "";
}
else {
    // if there was no operand
    if (s1.equals("") || s2.equals(""))
        s1 = s;
    // else evaluate
    else {
        double te;

        // store the value in 1st
        if (s1.equals("+"))

```

```

        te = (Double.parseDouble(s0) +
Double.parseDouble(s2));
        else if (s1.equals("-"))
            te = (Double.parseDouble(s0) -
Double.parseDouble(s2));
        else if (s1.equals("/"))
            te = (Double.parseDouble(s0) /
Double.parseDouble(s2));
        else
            te = (Double.parseDouble(s0) *
Double.parseDouble(s2));

        // convert it to string
        s0 = Double.toString(te);

        // place the operator
        s1 = s;

        // make the operand blank
        s2 = "";
    }

    // set the value of text
    l.setText(s0 + s1 + s2);
}
}
}

```